Heritability

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1 REML-based heritability estimation

These derivations are based on the Methods of Yang et al. [1].

1.1 Phenotype model

We can define a quantitative phenotype vector $\mathbf{y}$ as:

$$\mathbf{y} = \mathbf{X}_c\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

Where:

- $\mathbf{y} \in \mathbb{R}^N$: phenotypes. Centered so that $\mathrm{E}[\mathbf{y}] = 0$.
  - N: number of samples.
- $\mathbf{X}_c \in \mathbb{R}^{N\times M_c}$: normalized genotypes for causal variants.
  - M_c: number of causal variants.
  - Normalized according to $\mathbf{X}_{c,i} = \frac{\mathbf{X}'_{c,i}-2f_i}{\sqrt{2f_i(1-f_i)}}$.
    - $\mathbf{X}'_c \in \{0,1,2\}^{N\times M_c}$: allele dosages.
    - f_i: true population allele frequency for variant i.
    - Such that for each row (variant), $\mathrm{E}[\mathbf{X}_{c,i}] = 0$ and $\mathrm{Var}[\mathbf{X}_{c,i}] = 1$.
- $\boldsymbol{\beta} \in \mathbb{R}^{M_c}$: per-normalized-genotype causal effects.
  - Assume infinitesimal model.
  - Drawn from $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \sigma^2_\beta\mathbf{I})$.
    - σ^2_β: variance of causal effects.
    - $\mathbf{I} \in \mathbb{R}^{M_c\times M_c}$: identity matrix.
- $\boldsymbol{\epsilon} \in \mathbb{R}^N$: residual effects (i.e. error or noise term).
  - Drawn from $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2_\epsilon\mathbf{I})$.
    - σ^2_ϵ: residual variance.
    - $\mathbf{I} \in \mathbb{R}^{N\times N}$: identity matrix.

We assume $\mathbf{X}_c$, $\boldsymbol{\beta}$, and $\boldsymbol{\epsilon}$ are all independent from each other.

1.2 Variance of the phenotype

By making use of the independence between terms, we can define the variance-covariance matrix of $\mathbf{y}$ as:

$$\begin{aligned}
\mathrm{Var}[\mathbf{y}] &= \mathrm{Var}[\mathbf{X}_c\boldsymbol{\beta} + \boldsymbol{\epsilon}] \\
&= \mathrm{Var}[\mathbf{X}_c\boldsymbol{\beta}] + \mathrm{Var}[\boldsymbol{\epsilon}] \\
&= \mathrm{Var}[\mathbf{X}_c]\mathrm{Var}[\boldsymbol{\beta}] + \mathrm{Var}[\boldsymbol{\epsilon}] \\
&= \sigma_\beta^2(\mathbf{X}_c\mathbf{X}_c^\intercal) + \sigma_\epsilon^2\mathbf{I} \\
&= M_c\sigma_\beta^2\frac{\mathbf{X}_c\mathbf{X}_c^\intercal}{M_c} + \sigma_\epsilon^2\mathbf{I} \\
&= \sigma_g^2\mathbf{G} + \sigma_\epsilon^2\mathbf{I}
\end{aligned}$$

where we define $\mathbf{G} = \frac{\mathbf{X}_c\mathbf{X}_c^\intercal}{M_c}$, with $\mathbf{G} \in \mathbb{R}^{N\times N}$, as the genetic relationship matrix (GRM) between individuals. The G_{ii} element is the variance of individual i's normalized genotype vector, while the G_{ij} element is the covariance of individuals i and j's normalized genotype vectors.

We also define $\sigma_g^2 = M_c\sigma_\beta^2$ to be the variance of the total additive genetic effects on the phenotype. We can therefore think of individuals' phenotypes as being derived from the sum of a genetic random effect $\mathbf{g} = \mathbf{X}_c\boldsymbol{\beta}$ and a residual random effect $\boldsymbol{\epsilon}$. From the general rule of multivariable statistics that $\mathrm{Var}[\mathbf{A}\mathbf{v}] = \mathbf{A}\mathrm{Var}[\mathbf{v}]\mathbf{A}^\intercal$, we can write the genetic random effect as coming from a normal distribution:

$$\mathbf{g} \sim \mathcal{N}(\mathrm{E}[\mathbf{X}\boldsymbol{\beta}], \mathrm{Var}[\mathbf{X}\boldsymbol{\beta}])$$

Because $\mathrm{E}[\boldsymbol{\beta}] = \mathbf{0}$, then $\mathrm{E}[\mathbf{X}\boldsymbol{\beta}] = \mathbf{0}$. As for the variance, we can reuse the definitions from earlier, $\mathrm{Var}[\mathbf{X}\boldsymbol{\beta}] = \mathbf{G}\sigma_g^2$. Therefore,

$$\mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{G}\sigma_g^2)$$

Narrow-sense heritability is defined as the proportion of phenotypic variance, σ_P^2, explained by additive genetic effects:

$$h^2 = \frac{\sigma_g^2}{\sigma_P^2} = \frac{\sigma_g^2}{\sigma_g^2 + \sigma_\epsilon^2}$$

1.3 Estimating the GRM

In practice, we likely do not know the exact set of causal variants and instead must estimate the GRM using a set of genotyped SNPs:

$$\mathbf{A} = \frac{\mathbf{X}\mathbf{X}^\intercal}{M}$$

Where $\mathbf{A}$ is the estimated GRM, $\mathbf{X}$ is the normalized genotype matrix of our genotyped SNPs, and M is the number of genotyped SNPs. Note that because we are also working with a sample, X is normalized using sample allele frequencies, $\mathbf{p}$:

$$\mathbf{X}_i = \frac{\mathbf{X}'_i - 2p_i}{\sqrt{2p_i(1-p_i)}}$$

However, this equation for $\mathbf{A}$ ignores the sampling error associated with each SNP. Let's consider the covariance computation between two individuals for SNP i, which is then summed across M SNPs to get the value for A_{jk}. When $j \neq k$:

$$\begin{aligned}
A_{ijk} &= x_{ij}x_{ik} \\
&= \frac{x'_{ij} - 2p_i}{\sqrt{2p_i(1-p_i)}} \frac{x'_{ik} - 2p_i}{\sqrt{2p_i(1-p_i)}} \\
&= \frac{(x'_{ij} - 2p_i)(x'_{ik} - 2p_i)}{2p_i(1-p_i)}
\end{aligned}$$

Because x'_{ij} and x'_{ik} are independent from each other, the expected value of this is:

$$\begin{aligned}
\mathrm{E}[A_{ijk}] &= \mathrm{E}[\frac{(x'_{ij} - 2p_i)(x'_{ik} - 2p_i)}{2p_i(1-p_i)}] \\
&= \frac{(\mathrm{E}[x'_{ij}] - 2p_i)(\mathrm{E}[x'_{ik}] - 2p_i)}{2p_i(1-p_i)} \\
&= \frac{(2p_i - 2p_i)(2p_i - 2p_i)}{2p_i(1-p_i)} \\
&= 0
\end{aligned}$$

This makes sense if our sample is of unrelated individuals. If the raw genotypes of two individuals at a SNP are independent from each other, then the covariance of their adjusted genotypes should also be zero. Furthermore, the variance of A_{ijk} is given by:

$$\begin{aligned}
\mathrm{Var}[A_{ijk}] &= \mathrm{Var}[\frac{(x'_{ij} - 2p_i)(x'_{ik} - 2p_i)}{2p_i(1-p_i)}] \\
&= \frac{\mathrm{Var}[(x'_{ij} - 2p_i)]\mathrm{Var}[(x'_{ik} - 2p_i)]}{(2p_i(1-p_i))^2} \\
&= \frac{\mathrm{Var}[(x'_{ij})]\mathrm{Var}[(x'_{ik})]}{(2p_i(1-p_i))^2} \\
&= \frac{(2p_i(2-p_i))(2p_i(2-p_i))}{(2p_i(1-p_i))^2} \\
&= 1
\end{aligned}$$

So, the variance in A_{jk} is independent of allele frequency, which is a desirable property.

But do these properties hold up when $j = k$ (i.e. variance of an indvidual's genotype)?

$$\begin{aligned}
A_{ijj} &= x_{ij}^2 \\
&= (\frac{x'_{ij} - 2p_i}{\sqrt{2p_i(1-p_i)}})^2 \\
&= \frac{(x'_{ij} - 2p_i)^2}{2p_i(1-p_i)}
\end{aligned}$$

For deriving $\mathrm{E}[A_{ijj}]$, we make use of:

$$\begin{aligned}
E[{x'}_{ij}^2] &= \sum_{n=0}^{2} \mathrm{E}[{x'}_{ij}^2 | {x'}_{ij} = n] \Pr({x'}_{ij} = n) \\
&= (0)^2(1-p_i)^2 + (1)^2 2p_i(1-p_i) + (2)^2(p_i)^2 \\
&= 2p_i - 2p_i^2 + 4p_i^2 \\
&= 2p_i^2 + 2p_i
\end{aligned}$$

Allowing us to calculate $\mathrm{E}[A_{ijj}]$:

$$\begin{aligned}
\mathrm{E}[A_{ijj}] &= \mathrm{E}[\frac{(x'_{ij} - 2p_i)^2}{2p_i(1-p_i)}] \\
&= \frac{\mathrm{E}[{x'}_{ij}^2 - 4p_i x'_{ij} + 4p_i^2]}{2p_i(1-p_i)} \\
&= \frac{\mathrm{E}[{x'}_{ij}^2] - 4p_i \mathrm{E}[x'_{ij}] + 4p_i^2}{2p_i(1-p_i)} \\
&= \frac{2p_i^2 + 2p_i - 4p_i(2p_i) + 4p_i^2}{2p_i(1-p_i)} \\
&= \frac{2p_i - 2p_i^2}{2p_i(1-p_i)} \\
&= \frac{2p_i(1-p_i)}{2p_i(1-p_i)} \\
&= 1
\end{aligned}$$

So, the expected variance of an individual's normalized genotype is 1 and independent of the allele frequency, which is a desirable property. However, this frequency-independence does not hold for the variance. For simplicity, let's denote $H = 2p_i(1-p_i)$ and make use of $\text{Var}[Y] = \text{E}[Y^2] - \text{E}[Y]^2$:

$$\begin{aligned}
\text{Var}[A_{ijj}] &= \text{Var}[\frac{(x'_{ij} - 2p_i)^2}{H}] \\
&= \frac{\text{Var}[(x'_{ij} - 2p_i)^2]}{(H)^2} \\
&= \frac{\text{E}[((x'_{ij} - 2p_i)^2)^2] - \text{E}[(x'_{ij} - 2p_i)^2)]^2}{(H)^2} \\
&= \frac{(H) - (H)^2}{(H)^2} \\
&= \frac{(H)(1 - H)}{(H)^2} \\
&= \frac{1 - H}{H} \\
&= \frac{1 - 2p_i(1 - p_i)}{2p_i(1 - p_i)}
\end{aligned}$$

The full derivation for why $\text{E}[((x'_{ij} - 2p_i)^2)^2] = \text{E}[(x'_{ij} - 2p_i)^2] = 2p_i(1 - p_i)$ is very lengthy algebraically, but can be shortcutted by using the formula for the higher moments of a binomially distributed variable, where $n = 2$ and $p = p_i$ [2]. Importantly, the variance of A_{jj} therefore depends on the allele frequencies of the SNPs, even after normalization. $\mathbf{A}$ would be a better estimator of the GRM if for the same individual, its variance did not depend on the allele frequency, so we want to adjust A_{ijj} so that $\text{E}[A_{ijj}] = 1$ like before, but also that $\text{Var}[A_{ijj}] = 1$ like for unrelated individuals.

To be frank, I am not sure how the authors derived it, but this equation holds both properties:

$$A_{ijj} = 1 + \frac{x'^2_{ij} - (1 + 2p_i)x_{ij} + 2p_i^2}{2p_i(1 - p_i)}$$

We now explicitly define $\mathbf{A}$ as follows:

$$A_{jk} = \frac{1}{M}\sum_{i=1}^{M} A_{ijk} = \begin{cases} \frac{1}{M}\sum_{i=1}^{M} \frac{(x'_{ij} - 2p_i)(x'_{ik} - 2p_i)}{2p_i(1 - p_i)}, & j \neq k \\ 1 + \frac{1}{M}\sum_{i=1}^{M} \frac{x'^2_{ij} - (1+2p_i)x'_{ij} + 2p_i^2}{2p_i(1 - p_i)}, & j = k \end{cases}$$

1.4 Estimating the genetic variance component through REML

Notice that earlier, we were able to define the phenotype $\mathbf{y}$ as a sum of random effects, treating $\mathbf{A}$ as a good approximation of $\mathbf{G}$:

$$\mathbf{y} = \mathbf{g} + \boldsymbol{\epsilon} \quad \text{where} \quad \mathbf{g} \sim \mathcal{N}(\mathbf{0}, \sigma_g^2\mathbf{A}) \quad \text{and} \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\epsilon^2\mathbf{I})$$

The unknown parameters here, σ_g^2 and σ_ϵ^2, can be estimated through REML (Restricted Maximum Likelihood; since this model doesn't include any fixed effects, simple Maximum Likelihood would also suffice). See the entry on "Maximum Likelihood" for a derivation of how variance component parameters are estimated through REML.

1.5 Heritability of a polygenic risk score

Here, we consider a polygenic score (PGS) $\hat{y}_i$ that acts to predict the trait value y_i of an individual. We assume the PGS only captures signal through the genetic component g_i of the trait and not through the random noise ϵ_i. That means we can model a polygenic score (PGS) as follows:

$$\hat{\mathbf{y}} = \mathbf{g} + \boldsymbol{\zeta}$$

where $\boldsymbol{\zeta} \in \mathbb{R}^N$ is random noise in the PGS: $\boldsymbol{\zeta} \sim \mathcal{N}(\mathbf{0}, \sigma_\zeta^2 \mathbf{I})$. PGS accuracy is commonly defined by its R^2 value:

$$\mathrm{R}^2 = \mathrm{Corr}[\mathbf{y}, \hat{\mathbf{y}}]^2$$

Under our assumptions, $\mathrm{R}^2 \leq h^2$. If the phenotype has been standardized so that $\sigma_P^2 = 1$, then $h^2 = \sigma_g^2$ and $\mathrm{R}^2 \leq \sigma_g^2$. We can derive the value of σ_ζ^2 as follows:

$$\begin{aligned}
\mathrm{R}^2 &= \mathrm{Corr}[\mathbf{y}, \hat{\mathbf{y}}]^2 \\
&= \frac{\mathrm{Cov}[\mathbf{y}, \hat{\mathbf{y}}]^2}{\mathrm{Var}[\mathbf{y}]\mathrm{Var}[\hat{\mathbf{y}}]} \\
&= \frac{\mathrm{Cov}[\mathbf{g} + \boldsymbol{\epsilon}, \mathbf{g} + \boldsymbol{\zeta}]^2}{\mathrm{Var}[\mathbf{g} + \boldsymbol{\epsilon}]\mathrm{Var}[\mathbf{g} + \boldsymbol{\zeta}]}
\end{aligned}$$

Because $\mathbf{g}$, $\boldsymbol{\epsilon}$, and $\boldsymbol{\zeta}$ are independent from each other:

$$\begin{aligned}
\mathrm{Cov}[\mathbf{g} + \boldsymbol{\epsilon}, \mathbf{g} + \boldsymbol{\zeta}] &= \mathrm{Cov}[\mathbf{g}, \mathbf{g}] + \mathrm{Cov}[\mathbf{g}, \boldsymbol{\zeta}] + \mathrm{Cov}[\boldsymbol{\epsilon}, \mathbf{g}] + \mathrm{Cov}[\boldsymbol{\epsilon}, \boldsymbol{\zeta}] \\
&= \mathrm{Var}[\mathbf{g}] \\
&= \sigma_g^2
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{Var}[\mathbf{y}] &= \mathrm{Var}[\mathbf{g} + \boldsymbol{\epsilon}] = \sigma_g^2 + \sigma_\epsilon^2 = \sigma_P^2 \\
\mathrm{Var}[\hat{\mathbf{y}}] &= \mathrm{Var}[\mathbf{g} + \boldsymbol{\zeta}] = \sigma_g^2 + \sigma_\zeta^2
\end{aligned}$$

Therefore,

$$\begin{aligned}
\mathrm{R}^2 &= \frac{(\sigma_g^2)^2}{(\sigma_g^2 + \sigma_\epsilon^2)(\sigma_g^2 + \sigma_\zeta^2)} \\
&= \frac{(\sigma_g^2)^2}{\sigma_P^2(\sigma_g^2 + \sigma_\zeta^2)}
\end{aligned}$$

Solving this for σ_ζ^2:

$$\begin{aligned}
\mathrm{R}^2 \sigma_P^2 (\sigma_g^2 + \sigma_\zeta^2) &= (\sigma_g^2)^2 \\
\sigma_g^2 + \sigma_\zeta^2 &= \frac{(\sigma_g^2)^2}{\mathrm{R}^2 \sigma_P^2} \\
\sigma_\zeta^2 &= \frac{(\sigma_g^2)^2}{\mathrm{R}^2 \sigma_P^2} - \sigma_g^2 \\
\sigma_\zeta^2 &= \sigma_g^2 \left(\frac{\sigma_g^2}{\mathrm{R}^2 \sigma_P^2} - 1 \right)
\end{aligned}$$

If we are working with a standardized phenotype, then $\sigma_P^2 = 1$ and this simplifies to:

$$\sigma_\zeta^2 = \sigma_g^2 \left(\frac{\sigma_g^2}{\mathrm{R}^2} - 1 \right)$$

This also makes clear why R^2 cannot be larger than the heritability under this model. If $\sigma_\zeta^2 = 0$, the PGS is exactly the genetic component, so $\hat{\mathbf{y}} = \mathbf{g}$ and $\mathrm{R}^2 = h^2$. Adding PGS noise increases $\mathrm{Var}[\hat{\mathbf{y}}]$ without increasing $\mathrm{Cov}[\mathbf{y}, \hat{\mathbf{y}}]$, so R^2 decreases. This also means that:

$$\mathrm{Var}[\hat{\mathbf{y}}] = \frac{(\sigma_g^2)^2}{\mathrm{R}^2 \sigma_P^2} = \frac{\sigma_g^2 h^2}{\mathrm{R}^2}$$

Suppose that the PGS is then scaled $\hat{\mathbf{y}}' = c\hat{\mathbf{y}}$ such that $\mathrm{Var}[\hat{\mathbf{y}}'] = 1$, where:

$$c^2 = \frac{1}{\mathrm{Var}[\hat{\mathbf{y}}]} = \frac{\mathrm{R}^2 \sigma_P^2}{(\sigma_g^2)^2}$$

We can then write the scaled PGS as:

$$\begin{aligned} \hat{\mathbf{y}}' &= c(\mathbf{g} + \boldsymbol{\zeta}) \\ &= c\mathbf{g} + c\boldsymbol{\zeta} \end{aligned}$$

The variance explained by the genetic component in the scaled PGS is:

$$\begin{aligned} \mathrm{Var}[c\mathbf{g}] &= c^2 \mathrm{Var}[\mathbf{g}] \\ &= \frac{\mathrm{R}^2 \sigma_P^2}{(\sigma_g^2)^2} \sigma_g^2 \\ &= \frac{\mathrm{R}^2 \sigma_P^2}{\sigma_g^2} \\ &= \frac{\mathrm{R}^2}{h^2} \end{aligned}$$

and the variance explained by the residual component in the scaled PGS is:

$$\begin{aligned} \mathrm{Var}[c\boldsymbol{\zeta}] &= c^2 \mathrm{Var}[\boldsymbol{\zeta}] \\ &= \frac{\mathrm{R}^2 \sigma_P^2}{(\sigma_g^2)^2} \left[\frac{(\sigma_g^2)^2}{\mathrm{R}^2 \sigma_P^2} - \sigma_g^2 \right] \\ &= 1 - \frac{\mathrm{R}^2 \sigma_P^2}{\sigma_g^2} \\ &= 1 - \frac{\mathrm{R}^2}{h^2} \end{aligned}$$

Therefore, if we fit the following REML model and treat $\mathbf{A}$ as a good approximation of the true GRM:

$$\hat{\mathbf{y}}' \sim \mathcal{N}(\mathbf{0}, \sigma_1^2 \mathbf{A} + \sigma_2^2 \mathbf{I})$$

the expected values of σ_1^2 and σ_2^2 are:

$$\begin{aligned} \mathrm{E}[\sigma_1^2] &= \frac{\mathrm{R}^2 \sigma_P^2}{\sigma_g^2} = \frac{\mathrm{R}^2}{h^2} \\ \mathrm{E}[\sigma_2^2] &= 1 - \frac{\mathrm{R}^2 \sigma_P^2}{\sigma_g^2} = 1 - \frac{\mathrm{R}^2}{h^2} \end{aligned}$$

In other words, the heritability of the scaled PGS is the ratio between PGS accuracy and trait heritability.

References

[1] Jian Yang, Beben Benyamin, Brian P. McEvoy, Scott Gordon, Anjali K. Henders, Dale R. Nyholt, Pamela A. Madden, Andrew C. Heath, Nicholas G. Martin, Grant W. Montgomery, Michael E. Goddard, and Peter M. Visscher. Common SNPs explain a large proportion of the heritability for human height. *Nature Genetics*, 42(7):565–569, 2010. doi: 10.1038/ng.608. URL https://www.nature.com/articles/ng.608.

[2] Wikipedia contributors. Binomial distribution, 2026. URL https://en.wikipedia.org/wiki/Binomial_distribution. Accessed 2026-05-03.